



Associated Prokaryotic Microbes in the Ctenidia and Mantle Tissue of Juvenile Giant Clam, *Tridacna gigas*

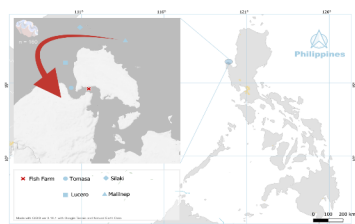


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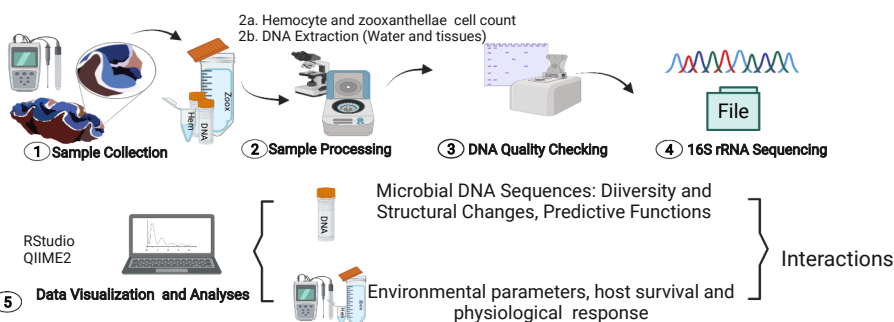
ABSTRACT

The microbiome has a significant influence on survival, health, and ecological functioning of the largest marine bivalve mollusk, the giant clam, *Tridacna gigas*. Microbial communities can be seeded from surrounding seawater and may accumulate in the giant clam's tissues. However, there is limited information on microbial community diversity in giant clam, their metabolic contribution to the giant clam host and their response to differing environmental conditions. In this study, we transplanted juvenile giant clams to sites with increasing proximity to fish farms and profiled bacterial composition in ctenidia (or gills) and mantle tissues using 16S rRNA sequencing. Results revealed clear separation between ctenidia, mantle, and seawater communities, which indicates community divergence despite direct contact with seawater. Members of Babeliales were exclusively detected in ctenidia tissues. At the site closest to fish farm where total suspended solids (data not shown) and nutrient concentration (i.e., Nitrate, Nitrite, and Ammonium) was highest, we observed greater relative abundance of Vibrionaceae in giant clam tissues, which correlated with greater host mortality. Our findings suggest that juvenile *T. gigas* harbor tissue-associated bacterial taxa that are responsive to changes in environment and that may influence giant clam adaptability.

METHODOLOGY

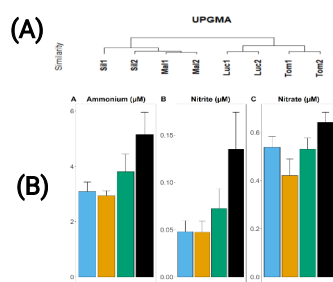


Transplantation sites with varying proximity to a fish farming zone

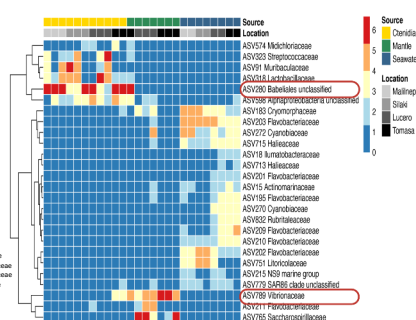
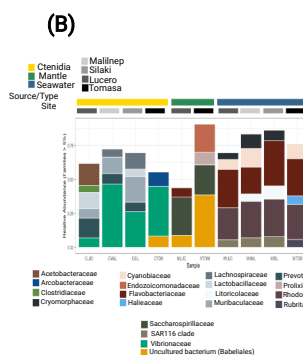
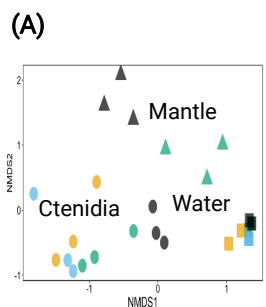


RESULTS AND DISCUSSION

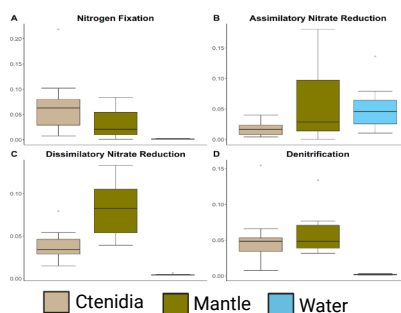
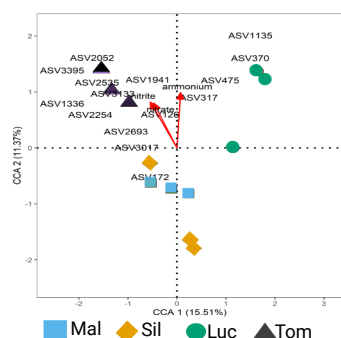
- Excess nutrient reaches beyond fish farming zone
- Clustering according to **sample types** is more prominent than clustering based on **site** as factor.
- Members of Babeliales and Vibrionaceae are relatively more abundant in Ctenidia and Mantle tissues, respectively.



far → near → mariculture site



- Presence of certain ASVs correlates with the environmental parameters (i.e. Ammonium, Nitrate, Nitrite)
- Microbial taxa associated with ctenidia and mantle tissues of juvenile giant clams may contribute to nitrogen cycling



As restocking efforts in the Philippines continue, this study has significant implications to understanding how giant clam populations may adapt to environmental perturbations they encounter at various sites, which is deemed crucial for conservation and management strategies

Future work incorporating robust functional analyses has to be done to further assess how prokaryotes contribute to and modulate host metabolism and health

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References: Caporaso, J.G., Lauber, C.L., Walters, W.A., Berg-lyons, D., Lozupone, C.A., Tumbaugh, P.J., Fierer, N., Knight, R., 2011. Global patterns of 16S rRNA diversity at a depth of millions of sequences per sample. *Proc. Natl. Acad. Sci. U. S. A.* 108, 4516–4522. <https://doi.org/10.1073/pnas.100080107>.
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